കണ്ണൂർ ജില്ലാ പഞ്ചായത്ത്

പൊതുവിദ്യാഭ്യാസ
വകുപ്പ്- കേരളം

ഡയറ്റ് കണ്ണൂർ

समग्र शिक्षा के रल
Samagra Shiksha Kerala
സമഗ്ര ശിക്ഷാ കേരളം



എസ് എസ് എൽ സി
വിദ്യാർത്ഥികൾക്കുള്ള
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ഫിസിക്സ്

ENGLISH MEDIUM

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2025-26

ആമുഖം

പ്രിയരേ,

കണ്ണൂർ ജില്ലാ പഞ്ചായത്തിന്റെ സമഗ്ര വിദ്യാഭ്യാസ പദ്ധതി, മുകളമായി തുടങ്ങി വിദ്യാർത്ഥി മനസ്സുകളിൽ പുഞ്ചിരിയായി വിടർന്നു കൊണ്ടിരിക്കുകയാണ്. SSLC വിദ്യാർത്ഥികൾക്ക് പാഠഭാഗങ്ങൾ സരളമായും സമഗ്രമായും മനസ്സിലാക്കാനും പൊതു പരീക്ഷകളിൽ ഉന്നത വിജയം നേടാൻ സഹായിക്കുന്നതിനും ആണ് SMILE മെറ്റീരിയൽ തയ്യാറാക്കുന്നത്. ഭൗതികശാസ്ത്രത്തിൽ മാറിയ പാഠപുസ്തകത്തിന്റെ ആദ്യ ഭാഗത്ത് നാല് അധ്യായങ്ങൾ ഉണ്ട്. ഓരോ അധ്യായത്തിലും പ്രധാന ആശയങ്ങൾ പ്രതിപാദിച്ച ശേഷം ചില മാതൃക ചോദ്യങ്ങൾ കൂടി ഉൾപ്പെടുത്തിയാണ് ഈ കൈപ്പുസ്തകം തയ്യാറാക്കിയിട്ടുള്ളത്. യൂണിറ്റ് ടെസ്റ്റുകൾക്ക് ആവശ്യമായിട്ടുള്ള മാതൃക ചോദ്യപേപ്പറുകൾ കൂടി ഉൾപ്പെടുത്തിയിട്ടുണ്ട്. അധ്യാപകർക്കും കുട്ടികൾക്കും ഒരുപോലെ പ്രയോജനപ്പെടുത്താൻ സാധിക്കുന്ന രീതിയിലാണ് ഇതിലെ വിവരങ്ങൾ ക്രമപ്പെടുത്തിയിട്ടുള്ളത്. മാറിയ പാഠപുസ്തകം ആയതുകൊണ്ട് തന്നെ ഇത് സമ്പൂർണ്ണമാണെന്ന് കരുതുന്നില്ല. വരും വർഷങ്ങളിൽ മെച്ചപ്പെടുത്താൻ സാധിക്കുന്ന തരത്തിൽ, ഈ കൈപ്പുസ്തകത്തെ അധ്യാപകർ ഉപയോഗിക്കുകയും ക്രിയാത്മകമായ നിർദ്ദേശങ്ങൾ നൽകുകയും ചെയ്യുമെന്ന പ്രതീക്ഷയോടെ....

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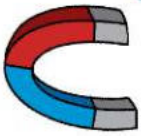
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1 Sound Waves

" This chapter discusses the propagation of sound and various phenomena, starting with oscillatory motion and moving through the various levels of wave motion."

Oscillation:

Oscillation is a periodic motion in which an object moves to and fro at regular intervals of time about its equilibrium position.

Period (T)

Period is the time taken for one oscillation.

Unit - **second(s)**

Frequency

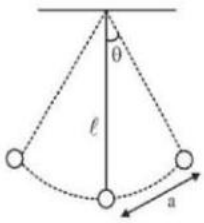
Frequency is the number of oscillations per second.

The unit of frequency is **Hertz. (Hz)**

Amplitude (a)

The magnitude of maximum displacement to one side from its equilibrium position is amplitude.

The SI unit of amplitude is **metre (m).**



A simple pendulum is a bob (weight) suspended from a string.

DON'T FORGET !

- *As the length of a simple pendulum increases, the period increases and the frequency decreases.*
  If "n" oscillations take "t" time,
- *Period, T = t/n second*
- *Frequency, f = n/t hertz*
- **Frequency (f) = $\frac{1}{\text{Period (T)}}$**

Model questions

1. An oscillation is.....

a) A movement from the equilibrium position to one side.

b) A movement from one side of the equilibrium position to the other.

c) A return to the starting position in the same direction.

2. What type of motion is the movement of a pendulum in a clock, the movement of a swing, etc.?

3. Which is the correct statement about the amplitude in oscillatory motion?

a) The distance moved from the equilibrium position to one side.

b) The maximum displacement from the equilibrium position in one direction.

c) The distance moved in both directions from the equilibrium position.

4. A pendulum that completes 20 oscillations in one minute...

a) What is the frequency?

b) What is the period?

5. A simple pendulum is a system in which a bob is suspended from a string.

a) How is the length of the string of the pendulum related to the frequency?

b) What is the change in the period when the frequency increases?

* Period, frequency, amplitude, etc. are also applicable to **oscillatory** motion.
* Natural vibration is the vibration that an object makes independently.
* **Natural frequency:** The natural frequency of an object is the frequency that occurs when it vibrates freely.
* A tuning fork is vibrating at its natural frequency,when vibrated freely
* A tuning fork marked 256 Hz will vibrate 256 times per second.
* The factors that affect the natural frequency of an object include the length, size, elasticity, and nature of the material.
* **Forced vibration:** Vibration caused by an excitation.
  When a mixer is running on a table, the table vibrates due to the induced vibration.
* **Resonance:** When the natural frequency of the object being excited and the natural frequency of the object being vibrated are the same, they are in resonance.
  An object undergoing resonance vibrates at its maximum amplitude.
* Examples of applications of Forced vibration and resonance
  # MRI scanning # Radio unit # In stethoscope # In megaphone horn
  # In musical instruments such as guitar, violin, veena, harmonium, mridangam, etc.



Model questions

1. When a tuning fork marked 512 Hz vibrates, the air around it also vibrates.

a) What does the marking on the tuning fork indicate?

b) What is the frequency of the air at this point?

c) What is the name of the phenomenon of air vibrating in this way?

2. Write two situations in which forced vibration is used.

3. A bucket of water, a PVC pipe and a tuning fork are given. Write an experiment in which resonance can be heard using these.

4. When a tuned tuning fork is placed at the mouth of a PVC pipe lowered into the water and the pipe is slowly raised, a louder sound is sometimes heard.

a) What is the name of the phenomenon of air vibrating inside the pipe?

b) What causes the loudness of the sound to increase?

c) What happens to the sound when the loudness increases?

d) Write the name of this phenomenon?

Wave motion

- Wave motion is the continuous propagation of energy received in a part of the medium through vibrations
- Some types of waves do not require a medium to travel. Such waves are called **electromagnetic waves.**

  Radio waves, infrared waves, visible light, X-rays, gamma rays, etc. are electromagnetic waves.
- Waves that require a medium to travel are called **mechanical waves.**

  Water waves, sound waves, seismic waves, etc. are mechanical waves.

Mechanical waves are transmitted in two ways.

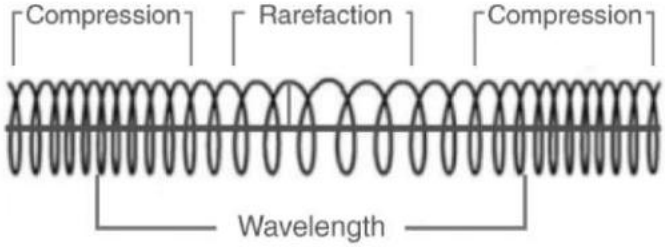
Longitudinal waves:

- The particles in the medium vibrate parallel to the direction of propagation of the wave.



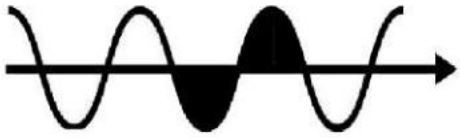
  If a slinky fixed at both ends is moved back and forth, the wave formed in it is longitudinal.
- **Sound** travels through a medium longitudinally.

  When a longitudinal wave travels, continuous pressure changes are formed in the medium.
- Regions of high pressure are called **compressions** and regions of low pressure are called **rarefactions**
- In a wave, **wavelength** is the distance between particles in the same state of vibration.
- In a longitudinal wave, wavelength is the distance between two adjacent **compressions** or between two adjacent **rarefactions.**



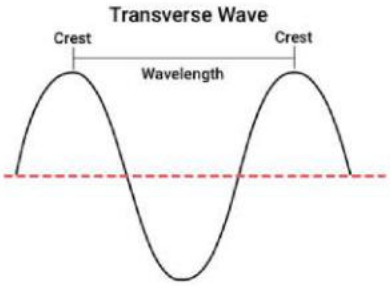
Transverse Wave

Transverse waves are waves in which the particles in the medium vibrate perpendicular to the direction of wave propagation.



- The part of the vibrating particles that is higher than the equilibrium position is called a **crest**, and the part that is lower is called a **trough**.
- Transverse waves are propagated as crests and troughs.

In a transverse wave, the wavelength is calculated as the distance between two adjacent crests or the distance between troughs.



Longitudunal waves	Transverse waves
☒ Particles vibrate parallel to the direction of wave propagation ☒ Compressions and rarefactions are formed ☒ Pressure variations occur in the medium	☒ Particles vibrate perpendicular to the direction of wave propagation ☒ Crests and troughs are formed ☒ No pressure variations occur in the medium



Model questions

1. What is wave motion?
2. Select the correct one from the statements examined

Sound requires a medium to travel

Sound is a mechanical wave

a) Only the first statement is correct

b) Only the second statement is correct

c) Both statements are correct

d) Both statements are incorrect

3. Waves that travel as pressure fields belong to which category?
4. All electromagnetic waves are .....waves
5. In a transverse wave, the part that rises above the equilibrium position is known as.....

Characteristics of waves

Amplitude, frequency, period, wavelength, wave speed

Amplitude : The maximum displacement of a particle in a wave motion from its equilibrium position to one side

Period: The time taken by a particle in the medium to complete one oscillation.

Frequency: If one oscillation of a particle in the medium is considered as a cycle, frequency is the number of cycles that pass through a point in one second.

Wavelength : In a wave, wavelength is the distance between particles in the same state of vibration. Wavelength is the distance that the wave travels in one second when the particle completes one oscillation.

Wave speed : The distance that the wave travels in one second is wave velocity



DON'T FORGET !

Wave speed = frequency x wavelength

$v = f \times \lambda$

When the velocity is constant, the frequency of the wave is inversely proportional to the wavelength.



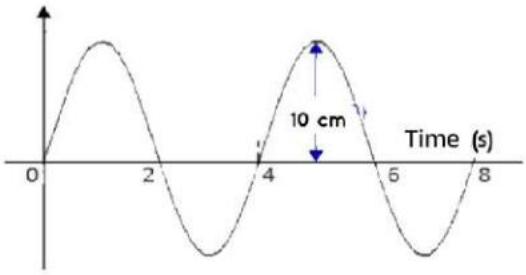
Model questions

1. Observe the picture

a) What is the length?

b) What is the frequency?

c) Find the speed of the wave if the distance between two adjacent peaks is 16 m



2. A sound wave has a frequency of 250 Hz. If its wavelength is 1.4 m, what is the speed of the wave?

3. Define wavelength in a longitudinal wave

4. Examine the statements about a wave with constant speed and choose the correct ones

a) As the frequency increases, the wavelength decreases

b) As the frequency increases, the wavelength increases

c) As the frequency increases, the wavelength does not change

d) Frequency and wavelength are inversely proportional.

Reflection of sound

Sound reflects off smooth surfaces.

Some materials are capable of absorbing sound.

Systems such as **sound boards and curved ceilings** take advantage of the reflection of sound. These help in good acoustics in a hall.

Repetitive reflection of sound

Repetitive reflection is the repetition of a sound that has been reflected once.

Echo

An echo is the hearing of its reflection after hearing the first sound.

The auditory experience of a sound remains in our ears for 1/10 of a second

This feature is **Persistance of audiability**

To hear an echo in air, there must be a distance of at least 17.5 m between the source and the reflection surface.

Reverberation is the sound produced by the repeated reflection of sound

Hearing Range

The lower limit of the frequency of sound that humans can hear is 20 Hz.

The upper limit of the frequency of sound that humans can hear is 20,000 Hz.

Sound waves with a frequency lower than 20 Hz are called **infrasonic**

Sound waves with a frequency higher than 20,000 Hz are called **ultrasonic**

Humans cannot hear the sound of the Galton whistle with a frequency of 30,000 Hz.

Bats move in the dark and catch prey with the help of ultrasonic sound

Uses of ultrasonic waves

- To break up kidney stones
- In physiotherapy
- To take images of internal organs such as kidney, liver, gall bladder and uterus.
- To clean spiral-shaped pipes
- To find the depth of water bodies with a device called SONAR

Seismic waves are waves that occur on the surface of the Earth during an earthquake.

The study of such waves is known as ***seismology***

Tsunamis *are huge waves that occur as a result of earthquakes that occur under the sea.*



Model questions

1. Write two uses of ultrasonic sound waves
2. Which scale measures the intensity of earthquakes?
3. If the speed of sound in water is considered to be 1500 m/s, can the echo of a sound emitted at a depth of 50 m be heard? Explain

MARK : 10 **UNIT TEST** TIME :30 Minutes

Chapter 1 : Sound Waves

Section A (1 score each)

1. Read the statements and find the correct conclusion and write it.

i. The time it takes for a pendulum to complete one oscillation is its period.

ii. A shorter pendulum takes longer to complete one oscillation than a longer pendulum

a) Both statements are true

b) Statement i is false and ii is true

c) Both statements are false

d) Statement i is true and ii is false.

2. Which of the following is not related to natural frequency?

a) Length of the object

b) Size of the object

c) Colour of the object

d) Elasticity of the object

3. Find and write the mechanical waves from the given

(Radio waves, sound waves, light waves, water waves)

Section B (2 scores each)

4. A simple pendulum completes 20 oscillations in a minute.

a) What is its frequency?

b) Calculate the period of the pendulum

5. Wave motion is the transmission of energy in the form of disturbance in a part of the medium.

a) What type of wave is a wave that travels in the medium as pressure fields?

b) What is the resulting displacement of a particle in the medium in wave motion?

Section C (3 marks each)

6. The frequency of the Galton whistle used to train dogs is 30000 Hz.

a) Can humans hear its sound?

b) What is the name of such sound waves?

c) Write any two uses of these



2 LENSES

"This lesson discusses the properties of lenses, image formation, and devices that use lenses."

- A lens is a transparent medium whose refractive surfaces are part of a sphere.
- Convex lenses focus light onto a point.

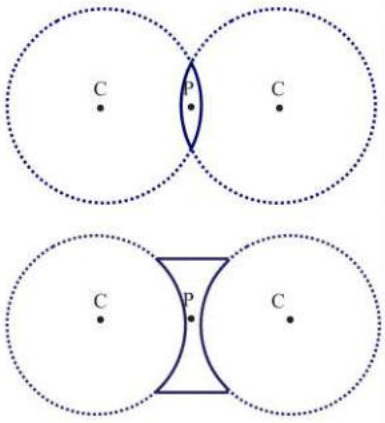


Convex Lens	Concave Lens
The middle part is thicker.	The middle part is thinner.
The edge is thin.	The edge is thick.
Concentrate light rays to a point	Unable to Concentrate light rays to a point

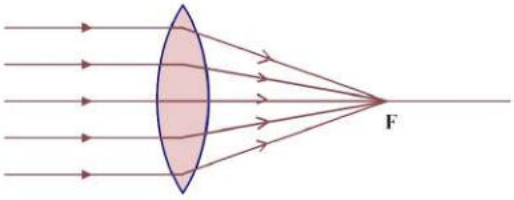


Terms related to Lenses

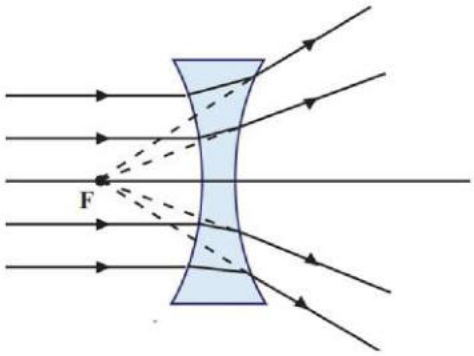
- **Optic centre :** The midpoint of a lens
- **Centres of Curvature**: The centres of the spheres which are the parts of a lens are centres of curvature.
- **Optic axis:** The imaginary line passing through the centres of curvature and the optic centre of a lens .
- **Aperture :** The area of the lens through which light passes is called aperture.



- **Principal focus of a convex lens:** Light rays incident on the lens near and parallel to the principal axis of a convex lens, after refraction, converge at a point on the optical axis on the other side of the lens. This is the principal focus of a convex lens.
- **Focal length :** The focal length is the distance from the optic center of the lens to the principal focus(f).



- **Principal focus of concave Lens:**
  Light rays, near and parallel to the optic axis incident on a concave lens, after refraction appear to diverge from a point on the optic axis on the same side of the lens.This point is the principal focus of concave lens (f)





DON'T FORGET !

- Images that can be projected on a screen are **real** images.Real images are **inverted** in nature.
- Images that cannot be projected on a screen are **virtual** images.Virtual images are **erect** in nature.

Image formation by a convex lens : Rules

1. A ray of light incident on a convex lens from a point parallel to the optical axis	After refraction through the lens, it passes through the principal focus on the other side.
2.A ray passing through the optical center.	Passes straight through without refraction
3. A ray of light that passes through the focus on the same side as the object and enters the lens.	After refraction, the light passes parallel to the axis.

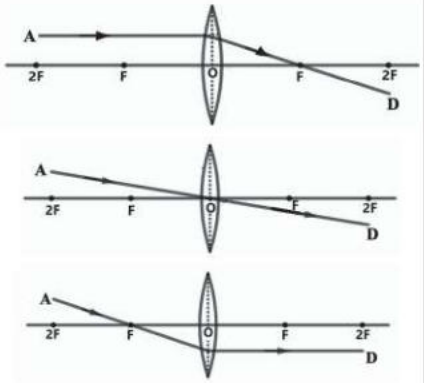
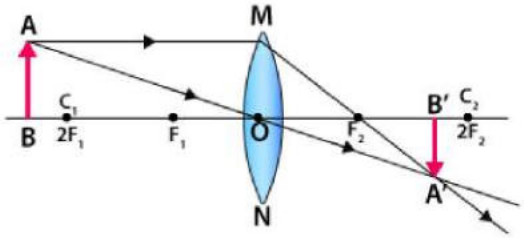


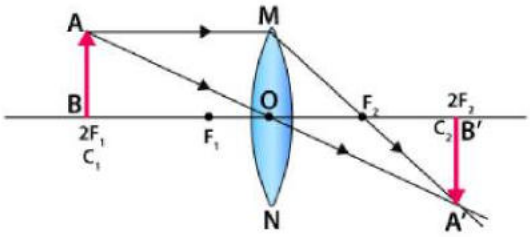
Image formation by a convex lens

Position of the object	Position of the image	Characteristics of image
Beyond 2F	Between F and 2F	Diminished, inverted, real
At 2 F	At 2F on the other side	Same size,inverted, real
Between2F and F	Beyond 2F	Magnified, inverted,real
At F	At infinity (Far away)	Magnified, inverted, real
Between F and lens	on the same side of the object	Magnified,erect, virtual

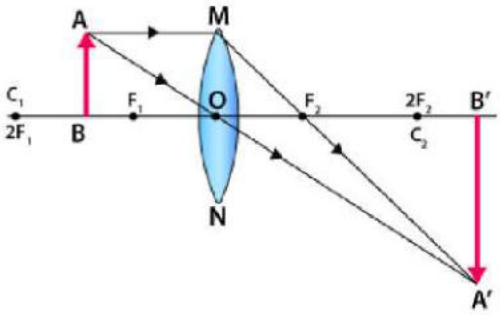
1. Beyond 2F



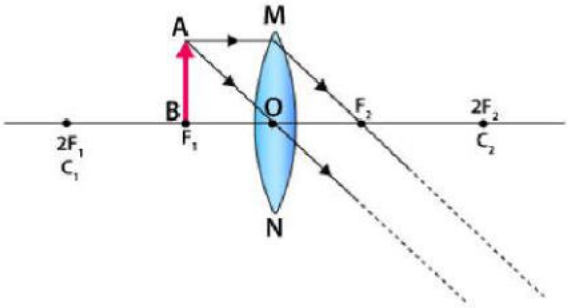
2. At 2F



3. Between F and 2F



4. At F



5.Between F and lens

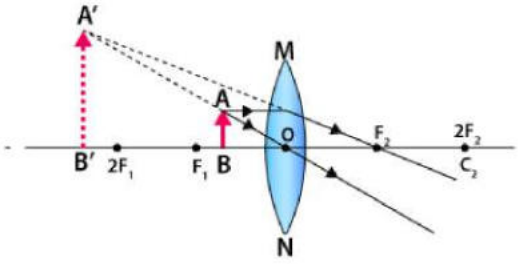
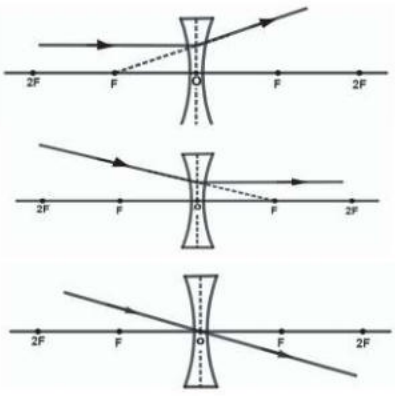


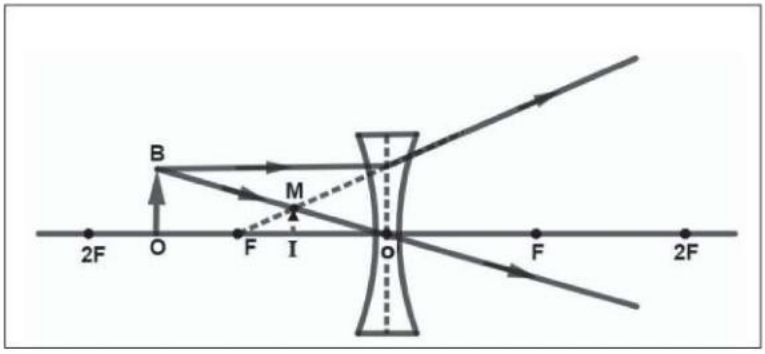
Image formation by a concave lens : Rules

A ray of light incident on a concave lens from a point parallel to the optical axis	After refraction, it appears to come from the focus on the same side.
2. Light rays passing through the lens towards the principal focus on the other side	Passes parallel to the optical axis.
3. Light rays passing through the optical center	It travels in a straight line without refraction.



DON'T FORGET !

- Since a concave lens refracts light rays, the image it forms will always be virtual.
- The image formed by a concave lens is always located on the same side of the object, between F and the lens.



Model questions

1. Write an experiment to identify a normal glass sheet and a convex lens.
2. List the difference between a convex lens and a concave lens.
3. A transparent medium in which each refracting surface is part of the sphere is known as —------.
4. What are the refractive surfaces of a lens?
   (spheres, circles, triangular prisms, squares)
5. Complete the table given below.

Position of Object	Position of image	Characteristics of the image
Beyond 2F	Between F and 2F	a
At 2F	b	c
Between F and 2F	d	e
At F	At infinity/Faraway	Magnified, inverted,real.
Between Fand Lens	e	f

6. The focal length of a convex lens is 10cm. If a piece of paper is to be burned by focusing sunlight, how far away should the piece of paper be from the lens?
7. Some statements related to convex lenses are given.Choose the correct statements.

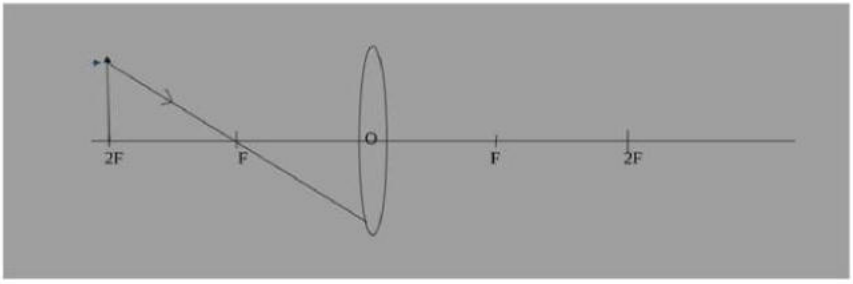
a. The edges are thicker and the middle is thinner.
b. The edge is thinner and the middle is thicker.

c. Focuses light rays parallel to the optical axis at a point on the optical axis.

d. Light rays parallel to the optical axis appear to diverge from a point on the optical axis.

i) a,b ii) a ,c iii) b, c iv) c, d

6.Complete the picture given below. Write the characteristics of the image.



Cartesian Sign Convention : General rules:

- 1. All distances should be measured from the optical center of the lens.
- 2. Distances measured in the same direction as the incident ray should be considered positive and those measured in the opposite direction should be considered negative
- 3. Measurements above the optical axis should be considered positive and those below should be considered negative.

Lens equation

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

f = focal length
u = object distance
v = image distance

$$f = \frac{uv}{u - v} \qquad u = \frac{fv}{f - v} \qquad v = \frac{uf}{u + f}$$



DON'T FORGET !

Convex lens	Concave lens
f - Positive u -Negative v-Positive (Real) or Negative (Virtual)	f - Negative u -Negative v -Negative (Virtual)

Magnification

- Magnification is the ratio of the height of the image to the height of the object.
- There is no unit for magnification, since it is a ratio of same physical quantities

$$m = \frac{v}{u} = \frac{h_i}{h_o}$$

hi = height of the image

ho = height of the object

v = distance to the image

u = distance to the object

According to Cartesian notation, if the magnification is **positive,** the image will be upright, and if the magnification is negative, the image will be inverted.



Model questions

1.A convex lens forms a real image of an object. What happens to the distance of the image when the object is brought closer to the focus of the lens?

2. List the differences between real image and virtual image.

3. An object is placed 30 cm away from a lens of focal length +20cm.

a. Find the distance to the image.

b. What type of lens is this? Convex / concave

c. Write 2 characteristics of the image.

4.The focal length of a concave lens is 15 cm. An object is placed 10 cm away from the lens.

a. Write the characteristics of the image

b. Calculate the distance to the image

5. A lens forms an image 3 times the size of the object. If the distance to the object is 20cm, calculate the distance of the image.

The power of the lens

- The reciprocal of the focal length is called power.
- The unit of power is the **diopter (D).**

$$P = \frac{1}{f}$$

- One diopter is the power of a lens with a focal length of one meter.
- If the power is positive, then the lens will be convex.
- If the power is negative, it will be a concave lens.

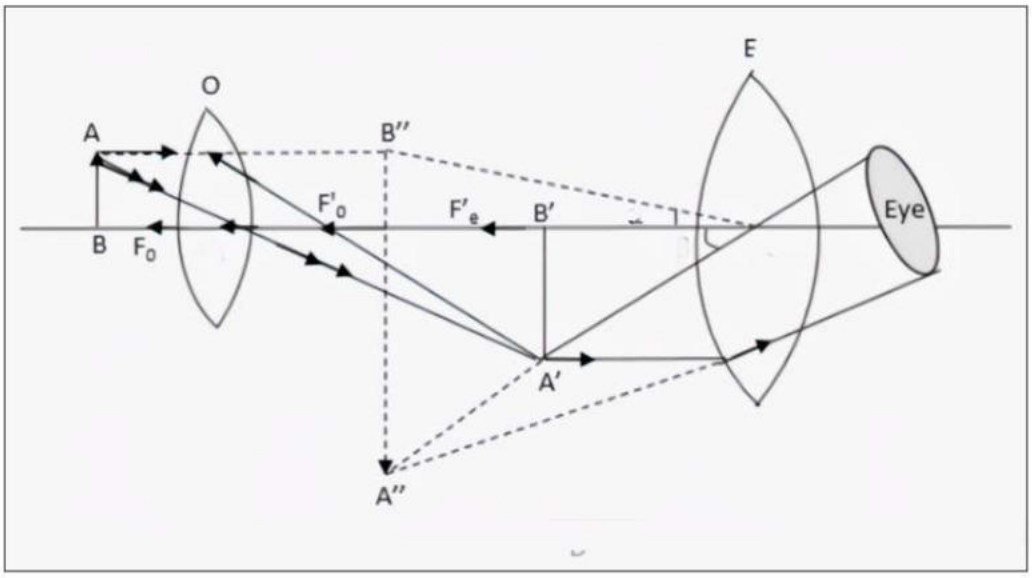
Compound microscope

Objective

- A lens placed near the object to be observed.

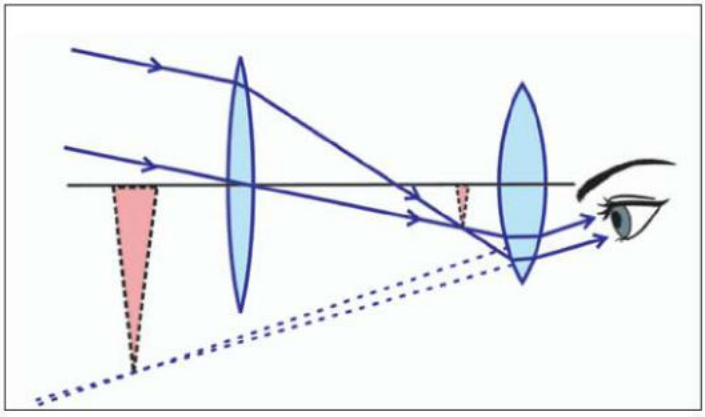
Eyepiece

- The eyepiece is the lens through which the image formed by the objective is observed.



Lens	Focal length	Position of Object	Position of image	Characteristics of image
Objective	Less	Between F and 2 F	Beyond 2 F	Real,inverted,magnified
Eyepiece	More	Between Lens and F	Same side	Virtual ,erect,magnified

Refracting Telescope



Lens	Focal length	Position of object	Position of Image	Characteristics of image
Objective	More	Far away	At F	Real, inverted , Small
Eyepiece	Less	Between Fand Lens	On the same side of the object	Erect, Bigger, Virtual



Model questions

1. The power of a convex lens is +2D. Find the focal length.
2. The table related to telescope is given below. Fill it in.

Lens	Focal length	Position of the object	Position of the image
Objective	a	Far away	b
Eyepiece	Less	c	d

3. The focal length of a concave lens is 20cm. What will be its power?

a.5D b. +5 D c. -0.5 D d. - 5D

4. An eye doctor's note says +2.00 D. What does this mean.....?
5. Why do they say not to look at the sun with a telescope?

MARK : 10 **UNIT TEST** TIME :30 Minutes

Chapter 2 : Lenses

1. what type of lens is thicker at the middle and thinner at edges.

a. Concave

b. Convex

c. Plane

d. none of the above **1 Mark**

2. The focal length of a convex lens is 10cm. An object is placed 20 cm away from the lens.

a.write any two characteristics of the image.

b.Calculate the magnification of the lens **2 Marks**

3. When an object is placed infront of a lens, a virtual, small and erect image is formed

a.identify the lens

b.Draw the figure of image formation. **3 Marks**

4.A convex lens forms a real image of an object. The image distance is 30cm and the magnification is 2 .

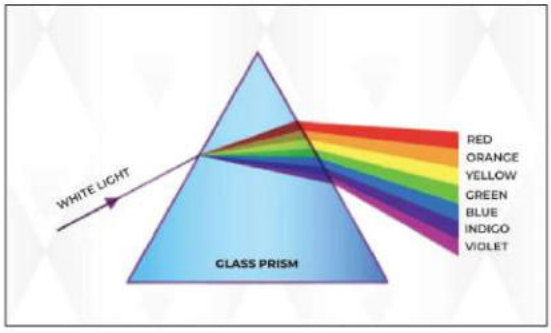
a.what is the object distance ?

b. Calculate the focal length of the lens **4 Marks**

3 The World of Colours and Vision

Dispersion of Light

- Dispersion of light is the phenomenon of splitting up of a composite light into its component colours.
- Light composed of different colours is a composite light.



The deviation of a ray of light due to refraction depends on the factors

1. Refractive index of the medium

2.Wavelength of the colour of light

The color with the shortest wavelength among the colours of light- Violet

The color with the longest wavelength among the colours of light – Red

- The shorter wavelength color (violet) has more deviation
- The longer wavelength color (red) has more deviation

Rainbow

A rainbow is formed as a result of the combined effects of refraction, dispersion and internal reflection of sunlight on water droplets in the atmosphere.

- A ray of sunlight passing through a water droplet undergoes refraction twice and internal reflection once.
- When the sun is in the east, the rainbow will be in the west.
- When the sun is in the west, the rainbow will be in the east.
- The colour found at the outer edge of the rainbow – Red
- The colour found at the inner edge of the rainbow – Violet



Model questions

1) The orderly arrangement of the component colors in white light is called —-------.

2) The phenomenon of splitting up of a composite light into its component colours is called ----------

3) Which of the following statements is/are correct regarding the dispersion of light?

I. The deviation of a ray of light depends only on the wavelength of the light.

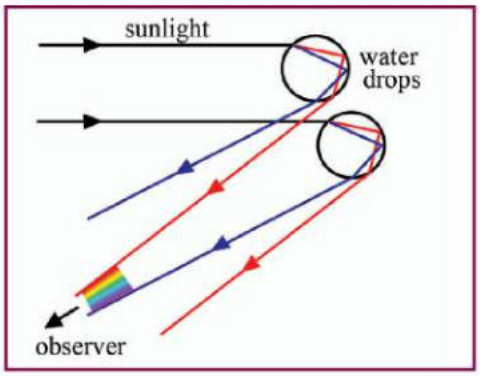
II.The shorter the wavelength, the more deviation occurs.

III. The deviation of a light ray depends on the refractive index of the medium and the wavelength of the light.

IV. The longer the wavelength, the less deviation occurs.

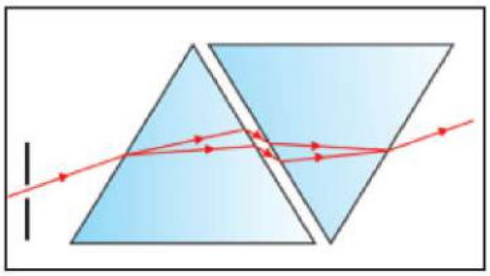
a) I,II b) II, III, IV c) I, III, IV d) only I

4) The image below shows light passing through water droplets.



a) When passing through water droplets where do light rays undergo refraction?
b) What happens to the refracted light rays inside the water droplets?
c) What color is seen at the outer edge of the rainbow?

5) Observe the picture and answer the questions given below



a) What is the cause of getting white light on the screen?
b) Which phenomena of light are involved here?

Electromagnetic Spectrum

The orderly distribution of electromagnetic radiations is known as the electromagnetic spectrum.

Electromagnetic radiations are given below in the ascending order of their wavelength.

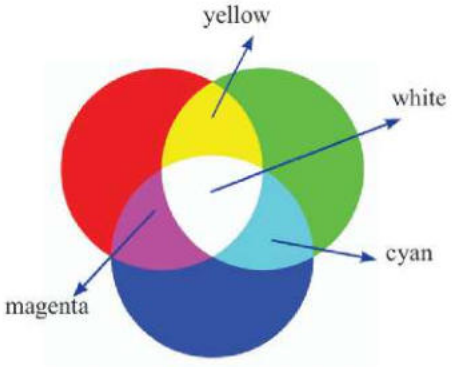
Gamma rays < X rays < Ultraviolet rays < Visible light < Infrared radiations < Radio waves

Features of Electromagnetic radiations

- They travel through vacuum at a speed of 300,000 kilometre per second (3×10^8 m/s).
- They do not require a medium to travel.
- The infrared radiation in the sunlight is the main reason for the heat in the Sun's rays.
- Ultraviolet radiation helps to produce vitamin D in our body.

Primary Colours and Secondary Colours

- Primary Colours – Red , Green , Blue
- The coloured light formed by combining any two primary colours is a secondary colour of light.
- When a secondary colour is combined with a primary colour, we get white light. Such pairs of colours are called complementary colours.
- Red + Green + Blue ⟶ White light
- Yellow + Magenta + Cyan ⟶ White light



Overlapping primary colours	Secondary Colour	Complementary Colour
Red + Green	Yellow	Blue
Red + Blue	Magenta	Green
Blue + Green	Cyan	Red

Persistence of Vision

The visual experience of that object persists for about 1/16 of a second in our eye.This phenomenon is the persistence of vision.

Examples :-

- When Newton's colour disc is rotated very fast,the combined effect of all the colours persists in our eyes and appears almost white.
- When a torch or a burning stick is rotated very fast, a ring of fire is seen.
- The rain looks like a glass rod.

Colour of Transparent Objects

- A filter in a primary color only transmits its own colour.
- A filter of secondary colour transmits light of its own colour and its component colours.

Light falling on the filter	Colour of the filter	Colour of light passing through the filter
White light	Red	Red
	Green	Green
	Blue	Blue
	Yellow	Yellow, Red , Green
	Magenta	Magenta, Red , Blue
	Cyan	Cyan, Blue , Green

Colour of opaque objects

- When sunlight falls on an opaque object, it reflects the colour of the object.
- An opaque object in a primary colour reflects light of its colour.
- An opaque object of a secondary colour can reflect light of its colour and its component colours.
- A surface that reflects all colours will appear white in white light.
- A surface that absorbs all colours appears dark.



Model questions

1)Which is the complementary colour of magenta?

a) Blue

b) Red

c) Green

d) Cyan

2) Various colour pairs are given.

I. Green,Red

II.Blue,Cyan

III. Magenta,Green

IV. Yellow,Blue

Which of these colour pairs produce white light?

a) I, III c) III ,IV

b) III, II d) II, IV

3) Fill in the blanks

a) Magenta + Green—> .....

b)..... + Yellow —> White light

c) Cyan +..... ..... —> White light

d) Blue +..... —> Magenta

4) If Newton's color disk is rotated rapidly, it appears white.

a) What peculiarity of the eye is this due to?

b) Write more examples related to this feature?

5) A child observes a red flower through a yellow filter.

a) What colour does the child see the flower as?

b) What color will you see if you observe a blue flower through the same filter?

6) Complete the table

Filter	Light falling on the filter	Transmits / Does not transmit
Cyan	Green	_
	_	Transmits blue
	White	_
	Yellow	_

8) Complete the table by writing the colour of the object given below in red, blue and magenta light.

Objects	Light	Colour of the object seen
Green Mango	Red	
	Blue	
	Magenta	
Red Apple	Red	
	Blue	
	Magenta	

Scattering of Light

Scattering is the irregular and partial directional deviation of light when it encounters particles in a medium.

- Colours with **shorter wavelengths** (violet, indigo, blue) have **more scattering.**
- Colours with **longer wavelengths** (red) have **less scattering.**
- As the particle size increases, the scattering increases.
- If the size of the particles is greater than the wavelength of light, the scattering will be the same for all colours.



DON'T FORGET !



Increasing Wavelength...
Decreasing Scattering....

Contexts related to scattering

- Blue Colour of the Sky.
- The sun appears red at sunrise and sunset.

• Tyndall Effect

• When light rays pass through a colloidal liquid or suspension, they get scattered, causing tiny particles to become illuminated, making the path of light visible.

• In winter, paths of light through the gaps of the branches of trees can be seen clearly due to scattering.



Model questions
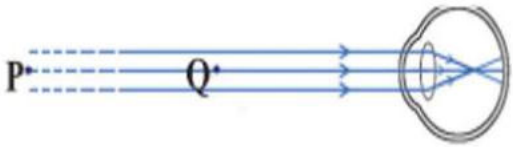
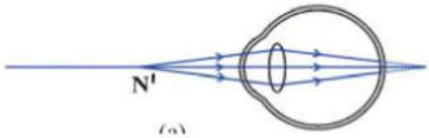
1) a)Which colour is scattered the most when sunlight passes through the atmosphere?
   b) How is the scattering of colours related to wavelength?
2) In winter, paths of light through the gaps of the branches of trees can be seen clearly due to scattering.
   a) What phenomenon is this related to?
   b) What light phenomenon is this related to?
   c) How the light phenomenon related to the size of the particles?
3) The colour of the sun appears red at sunrise and sunset. Which light phenomenon is this related to?

Eye and Vision

The power of accommodation of the eye

The ability of the eye to change the curvature of the lens and adjust the focal length so that the image of the object always falls on the retina,regardless of the position of the object, is the power of accommodation of the eye.

Near point	Far point
● The nearest point at which an object can be seen clearly. ● For healthy eyes, the minimum distance for clear vision is 25 cm.	● The farthest point at which an object can be seen clearly. ● For healthy eyes, this distance is considered to be infinity.

Defects of the eye		Reason for this defect	Remedy
Short sightedness / Myopia	can see nearby objects clearly but cannot see distant objects.	*The size of the eyeball is larger. *The power of the lens is more.	Use a concave lens with suitable power.
			
Long sightedness / Hypermetropia	can see distant objects clearly but cannot see nearby objects clearly.	* The size of the eyeball is smaller. *The power of the lens is less.	Use a convex lens with suitable power.
			
Presbyopia	For older people, the distance to the near point may be more than 25 cm.	Such people have less power of accommodation.	

Light Pollution

Light pollution refers to the creation of artificial light in excessive amounts and intensity.



Model questions

1) How far away is the near point of a person with Myopia ?

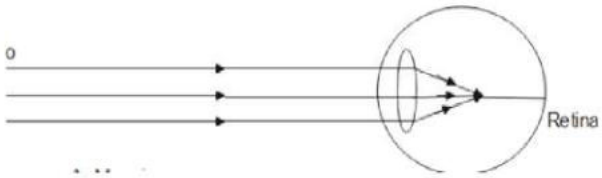
2) a) What are the difficulties faced by a person with long sightedness?

b) What is the reason for this?

c) How can this defect of the eye be rectified?

3) Prepare notes including facts to consider when preparing a seminar in class regarding the consequences of light pollution.

4) observe the figure



a) Which eye defect does the picture indicate?

b) What are the causes of this defect?

c) How can this defect of the eye be rectified?

5) Some statements are given below.

a)The near point will be more than 25 cm.

b)It can be solved using a convex lens of suitable power.

c)The power of the lens will be higher in people with this defect.

d)People with this defect have a small eyeball.

6. What is appropriate regarding long sightedness?

I) a , b , d

II) a , c, d

III) c, d

IV) only b

MARK : 10 **UNIT TEST** TIME :30 Minutes

Chapter 3 : The World of Colours and Vision

1) The coloured light formed by combining any two primary colour is a ..... (1)

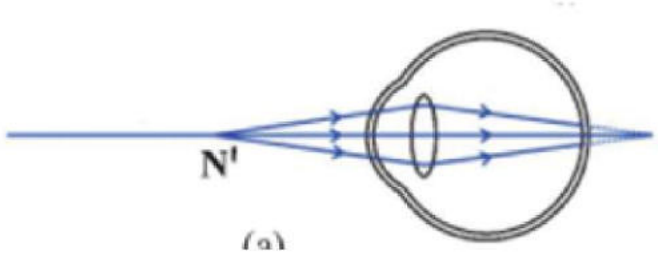
2) What is the scientific explanation for using red light for warning lamps? (2)

3)a) Write the radiations contained in the electromagnetic spectrum in descending order of their wavelength?

b) Which of these radiations causes heat?

c) Which of these radiations helps in producing vitamin D in the body? (3)

4) Observe the figure and write the questions given below



(a)

a)Where is the image formed in the figure?

b) Can people with such visual defects see an object at infinity?

c) Write the cause and solutions for this eye defect? (4)

4 Magnetic Effect of Electric Current

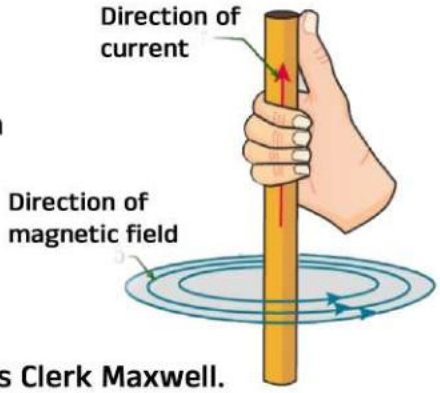
A magnet is an object that has always fascinated man. Electricity and magnetic force are inextricably linked. Electric current is the flow of electrons. Electrons are also behind magnetic force! There will be electric current in a coil that continuously crosses magnetic rings. There will be a magnetic ring around the coil through which electricity passes. Motors in the world are also built based on these principles. Dynamo and generator are all waves. Radiations, including light, electromagnetic radiation, light particles, time, are waves with the characteristics of electricity and magnetism. This lesson discusses the magnetic effect of electricity in detail

Electromagnetic Effect

A magnetic field is created around a straight conductor as a result of electric current. This was discovered by **Christian Oersted.**

Right-Hand Thumb Rule

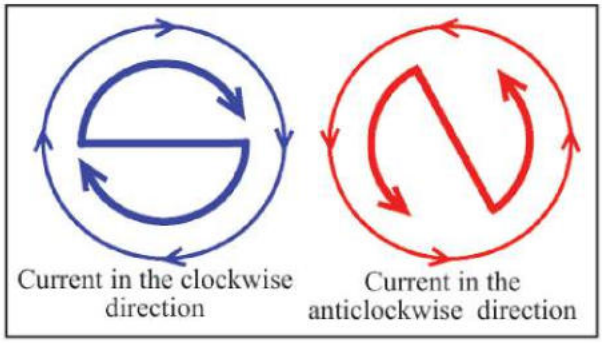
Imagine holding a conductor with your right hand in such a way that the thumb points in the direction of the **electric current**, the fingers curled around the conductor will indicate the direction of the **magnetic field.**



The right-hand thumb rule was developed by **James Clerk Maxwell.**
The right-hand thumb rule helps in understanding the direction of the **magnetic field** around a straight conductor carrying electricity.

DON'T FORGET !

Poles of Solenoid



Bar magnet	Current carrying solenoid
Magnetism is permanent	Magnetism is temporary
Magnetic strength cannot be varied.	Magnetic strength can be varied.
Polarity cannot be changed	Polarity can be changed

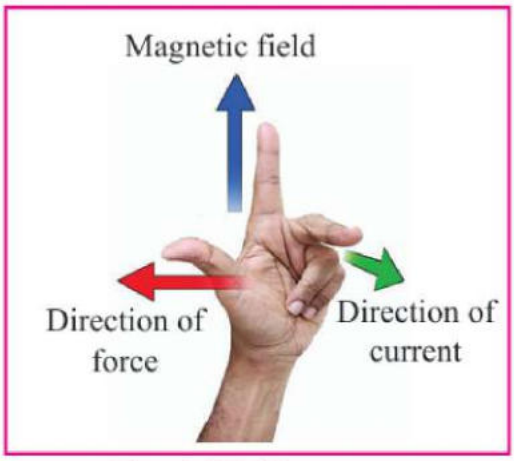
Fleming's Left Hand Rule

If we stretch the thumb, first finger, and second finger of your left hand so that they are all at right angles to each other, Then:

Thumb - Indicates the direction of force (motion)

First finger (fore finger) - Indicates the direction of magnetic field

Second finger (Middle finger)- Indicates the direction of current



Electric motor

Motor Principle

A current carrying conductor placed in a magnetic field experiences a force.

An electric motor is a device that changes electrical energy into mechanical energy

Working principle : ***motor principle***

Energy change : Electric energy ⟶ Mechanical energy

Parts

• Magnetic poles •Armature •Split rings •Graphite brushes

Moving Coil Loudspeaker

Working principle : **motor principle**

Energy change : Electric energy ⟶ Sound energy

Working

This electric signals (audio signals) from a microphone are first amplified using an amplifier.

↓

These signals are then sent through a voice coil.

↓

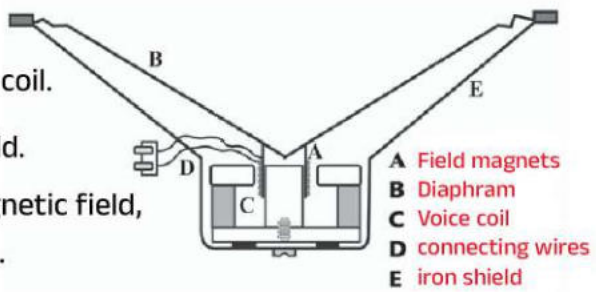
The voice coil is placed inside a magnetic field.

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Since the coil carries current and is in a magnetic field, It experiences a force and begins to vibrate..

↓

This vibration makes the diaphragm vibrate, which produces the sound we hear.





Model questions

1. Which of the following statements correctly describes the working of a loudspeaker?
- Sound energy is converted into electrical energy based on the motor principle.
- Electrical energy is converted into sound energy based on the motor principle.
- Electrical energy is converted into sound energy based on electro magnetic induction.
- Sound energy is converted into electrical energy based on electromagnetic induction.

2.In Right Hand Thumb Rule, the thumb indicates the direction of.....

3.Which of the following device works on motor principle?
(Generator, Transformer, Inductor, Moving coil loud speaker?

4. Find the relation in the first pair and complete the second.
(i) Electric iron : Heating effect
(ii) Electric motor : .....

5. Which of the following is not a part of moving coil loudspeaker?
(voice coil, diaphragm, permanent magnet, slip rings)

6. The prominent effect produced when a current passes through a solenoid is.....
(chemical effect, mechanical effect, magneti effect, light effect)

7. Name the arrangement in a DC motor to sustain the rotation of the armature in one direction.

8. Which part of a DC motor does not move?
(Armature, split rings, graphite brushes)

9. Select and write the device that works on the motor principle
(AC generator, moving coil microphone, DC generator, moving coil loudspeaker)

10. What magnetic polarity will the end of a solenoid exhibit when current flows in a anticlockwise direction.

11. Correct the following statement by changing the underlined words.
"The direction of deflection of magnetic needed placed near a current carrying conductor depend up on the intensity of electric current through that conductor".

12. Find odd one
(Armature, graphite brush, voice coil, split rings)

13. Explain the working of split ring commutator

14. Draw the magnetic field pattern around a vertical straight conductor carrying current downward.

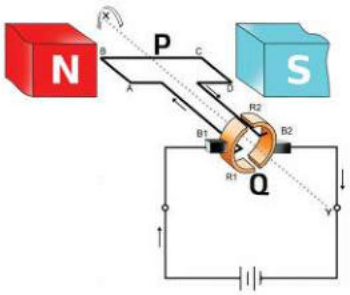
15. A conductor kept in a magnetic field experiences a force when current passes through it.
a) Name the rule that helps us to find the direction of the force.
b) Name any two devices that work based on this principle

16.Classify the following statements and write them down in the table suitably

- The magnetism is permanent.
- The polarity can be changed.
- The magnetism is temporary
- The polarity cannot be changed

Bar magnet	Current carrying Solenoid

17.Observe the schematic diagram of a device



a) Identify the device and write the energy change takes place in this device

b) Name the parts labelled P and Q

c) State the rule that helps to find the direction of motion of the part P when this device work

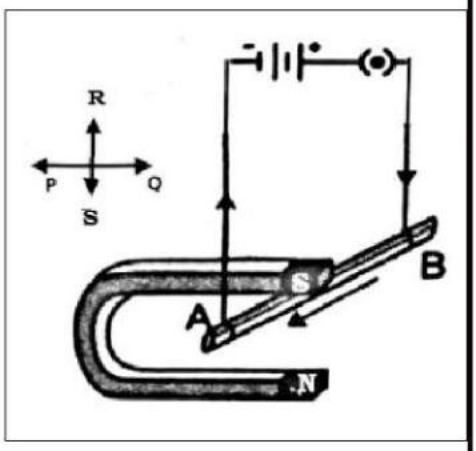
d) What is the function of the part 'Q' in the device?

18. DC flows through a solenoid.

a) What is the polarity at one end of the solenoid if the direction of current at that end is clockwise?

b) Suggest one method to interchange the magnetic strength of the solenoid carrying current.

19.A freely moving conductor A,B is in a magnetic field.

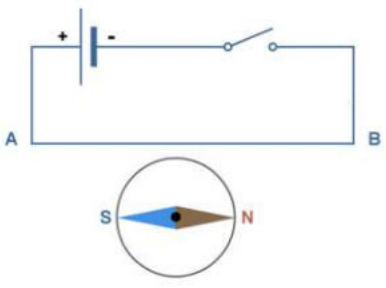


a) What happens when electricity flows through the conductor?

b) What is the reason for this?

c) What is observed if the direction of the current is reversed? What is the reason?

20.. A straight conductor AB is arranged parallel to a magnetic needle capable to rotate freely as shown in the figure.



a) Give the direction of deflection of the north pole of the magnetic needle when the circuit is switched on (clockwise / acticlockwise)

b) What is the reason for the deflection of the magnetic needle?

c) Suggest a method to reverse the direction of deflection of the magnetic needle.

MARK : 10 **UNIT TEST** TIME :30 Minutes

Chapter 4 : Magnetic effect of Electric current

1) From the given list identify the correct statements related to Fleming's left hand rule **(1)**

a) Thumb indicates the direction of force

b) Forefinger indicates direction of current

2) a) Write down the 2 main parts of a moving coil loudspeaker. **(1)**

b) Write the working principle of moving coil loudspeaker. **(1)**

3) Write 3 differences between bar magnet and solenoid. **(3)**

4) A conductor is hung freely in between the poles of a magnet

a) What happens to the conductor when electric current is passed through it?

b) Explain the reason behind this.

c) Which law helps you to explain this? **(4)**